Gating Preparation

File: Gating.kicad_sch

Gated Amp

File: Gated_amplifier.kicad_sch

Holdoff Trigger

File: Holdoff_Trigger.kicad_sch

+VDC

J101

Conn_01x03_Pin

1

2

3

GND_ref

−VDC

Shaping and Peak detection

File: Shaping_Peakdet.kicad_sch

SCA and Logic

File: SCA_Logic.kicad_sch

Counting

File: Counting.kicad_sch

Counting_pt2

File: counting_cont.kicad_sch

J102

R101

50

1

2

amp_out

J104

R102

50

1

2

logic_out

J106

R103

50

1

2

Gating Monitor

J103

1

2

DT_Reset

J105

1

2

Detector Signal

−VDC

C101 C102 C103 C104 C105

100u 100n 100n 1n 1n

GND_ref

+VDC

C106 C107 C108 C109 C110 C111 C112 C113 C114

100u 100n 100n 100n 100n 1n 1n 1n 1n

GND_ref

H101 H102 H103 H104

Mounting_Hole_Pad

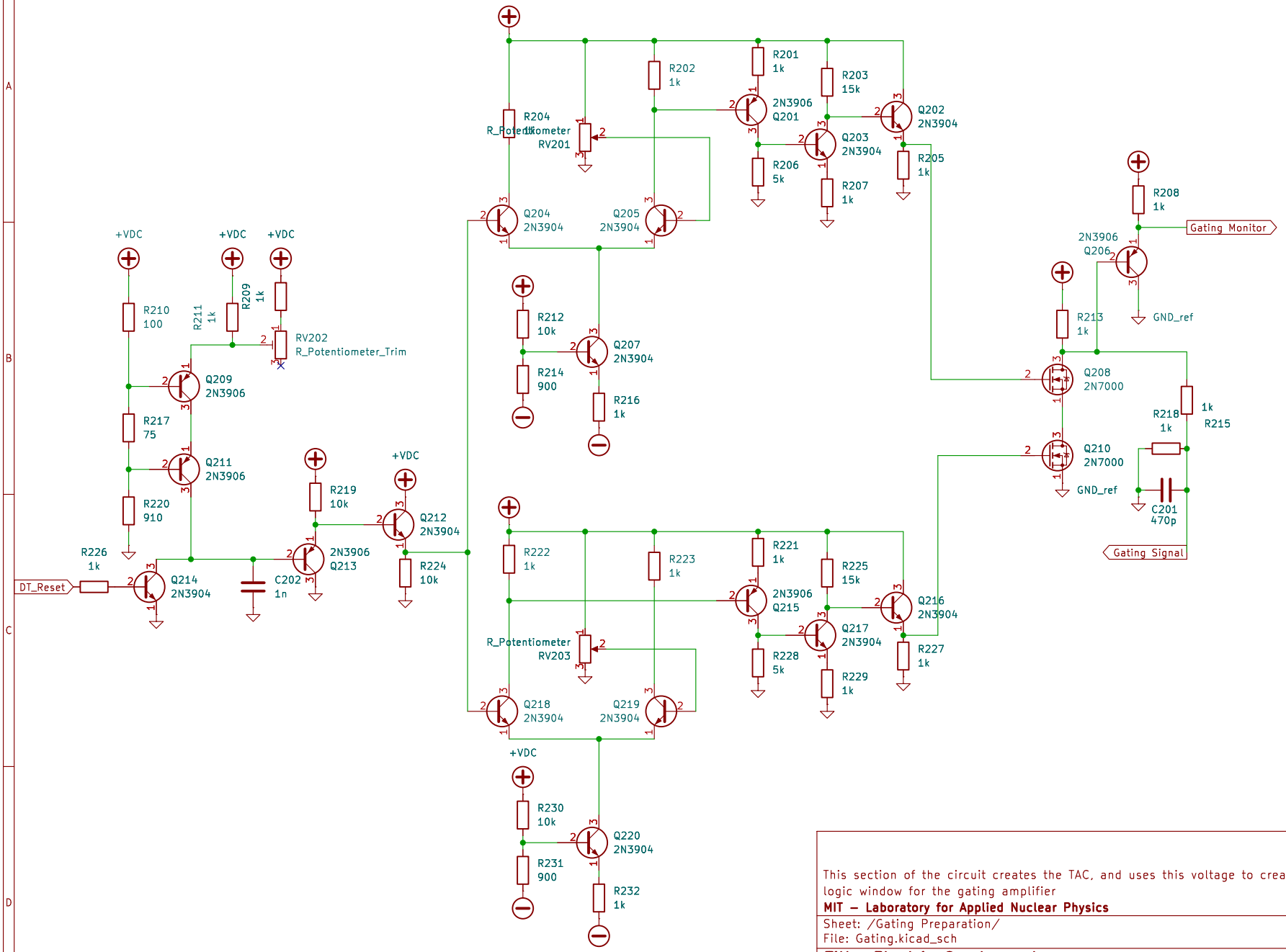
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Sheet: /

File: ECrypt_RevC.kicad_sch

Title: Electric Cryptography

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This section of the circuit creates the TAC, and uses this voltage to create a logic window for the gating amplifier

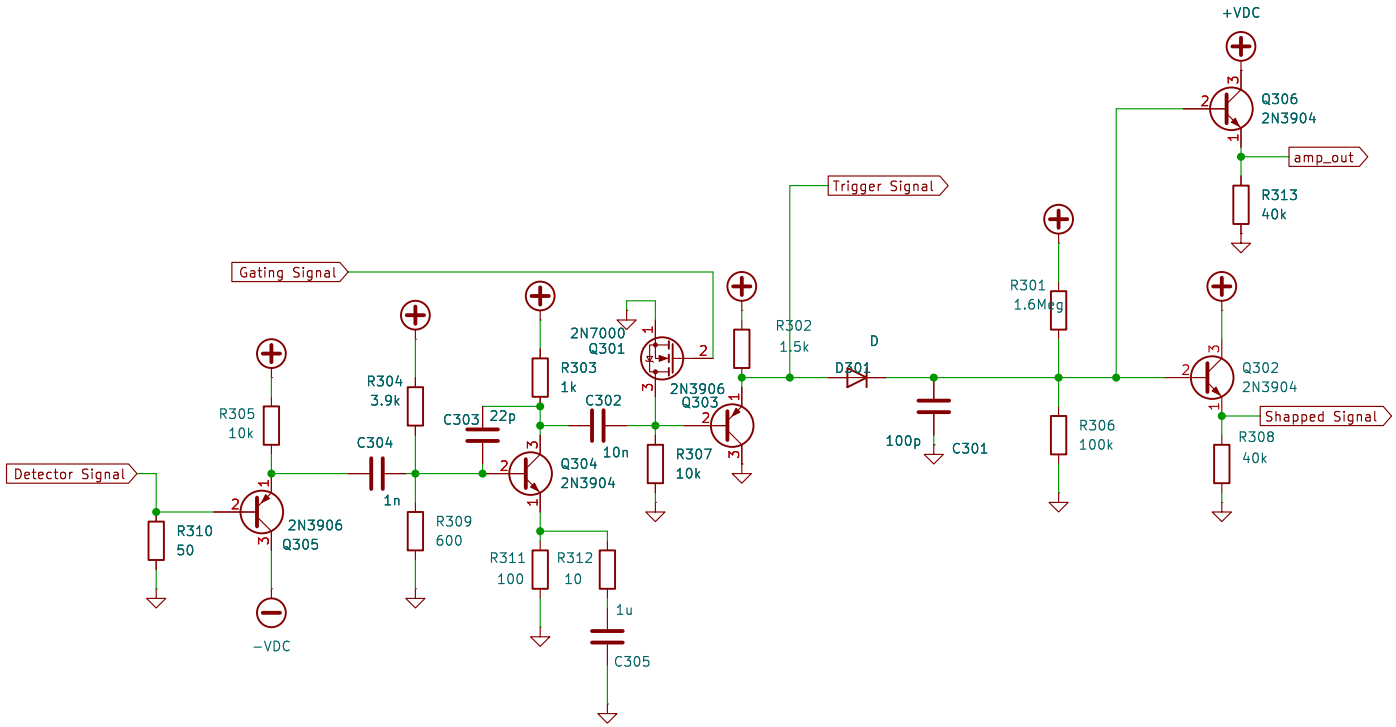
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Sheet: /Gating Preparation/

File: Gating.kicad_sch

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The gated amplifier, which amplifies pulses ONLY when the gating signal is low

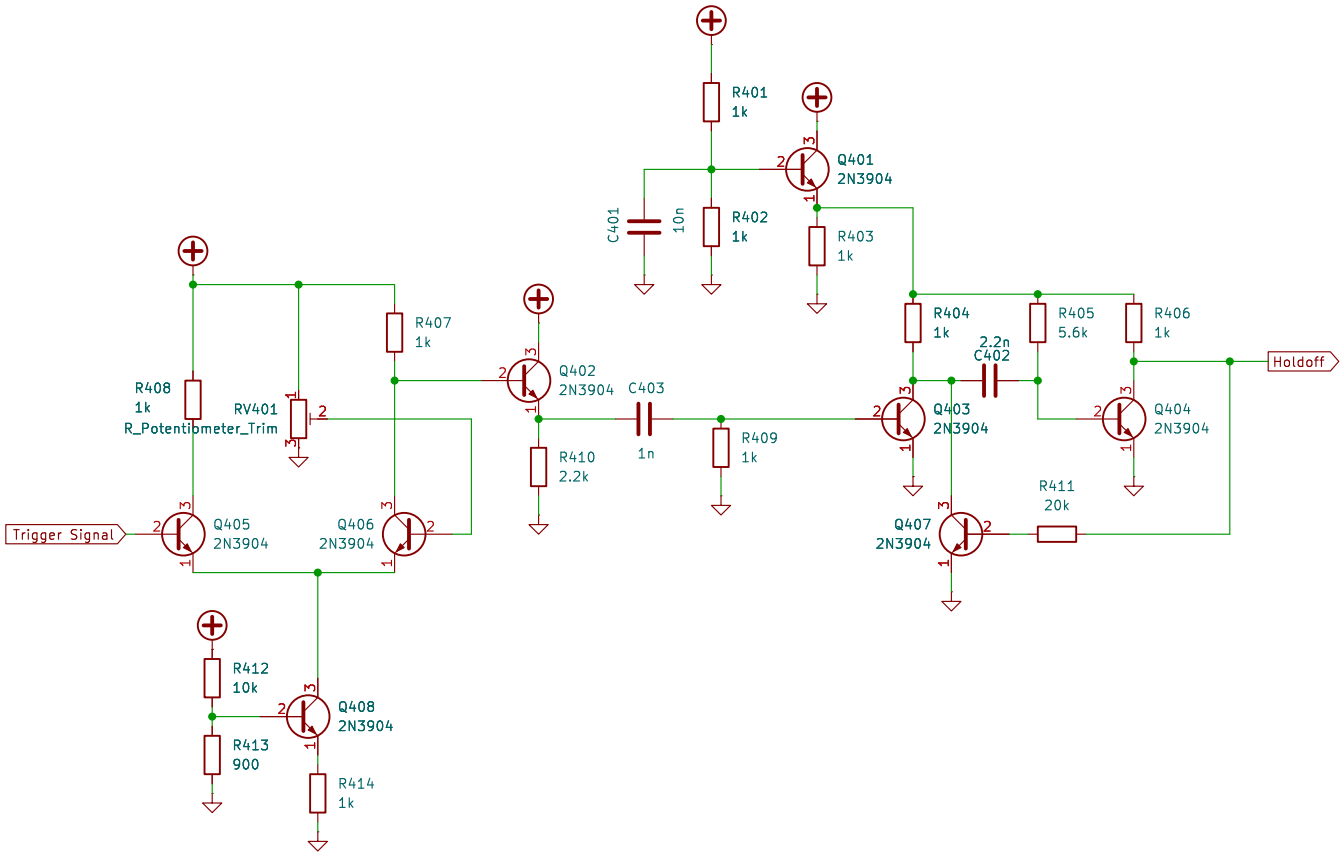
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Sheet: /Gated Amp/

File: Gated_amplifier.kicad_sch

Title: Electric Cryptography

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The holdoff creates a pulse triggered by the rising edge of the signal, and prevents the SCA from immediately when the LLD is crossed

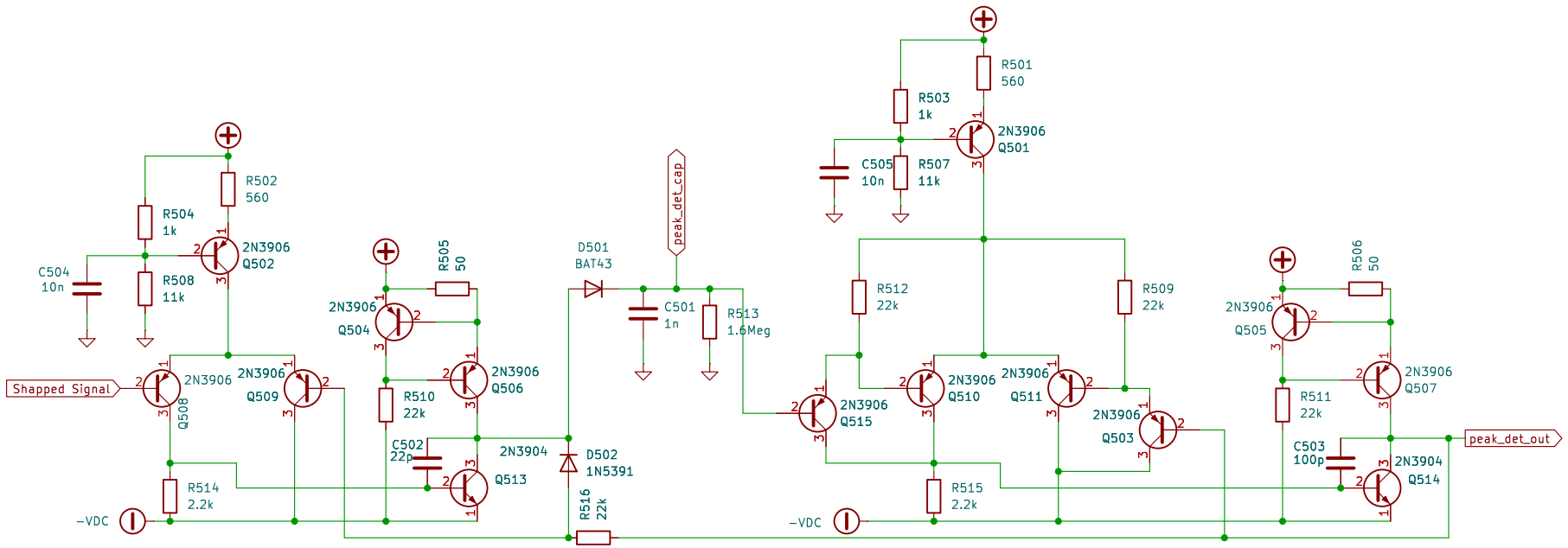
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Sheet: /Holdoff Trigger/

File: Holdoff_Trigger.kicad_sch

Title: Electric Cryptography

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Peak detector which captures the peak amplitude of each pulse. It is this peak which is compared to the LLD and ULD to fire the SCA

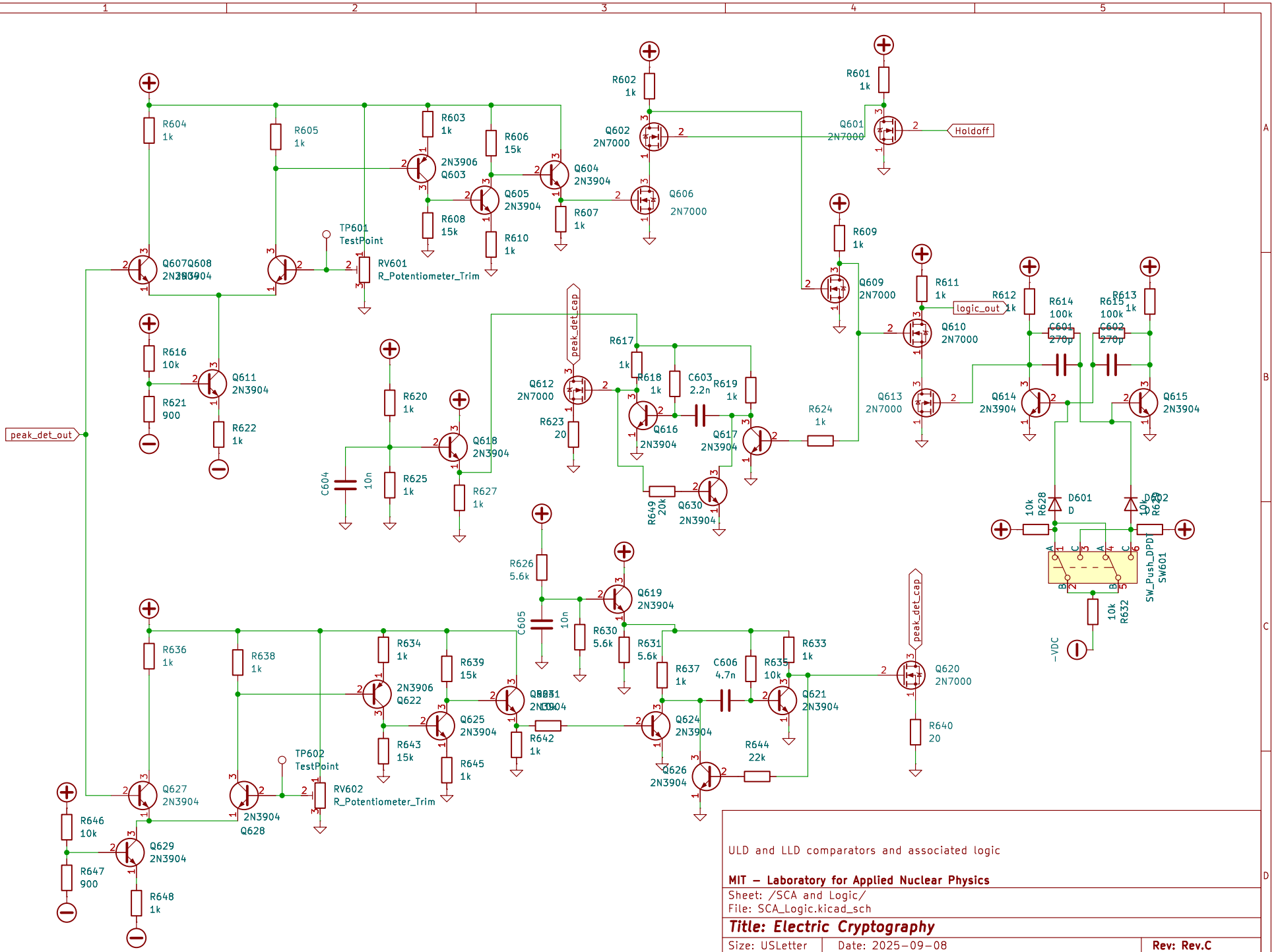
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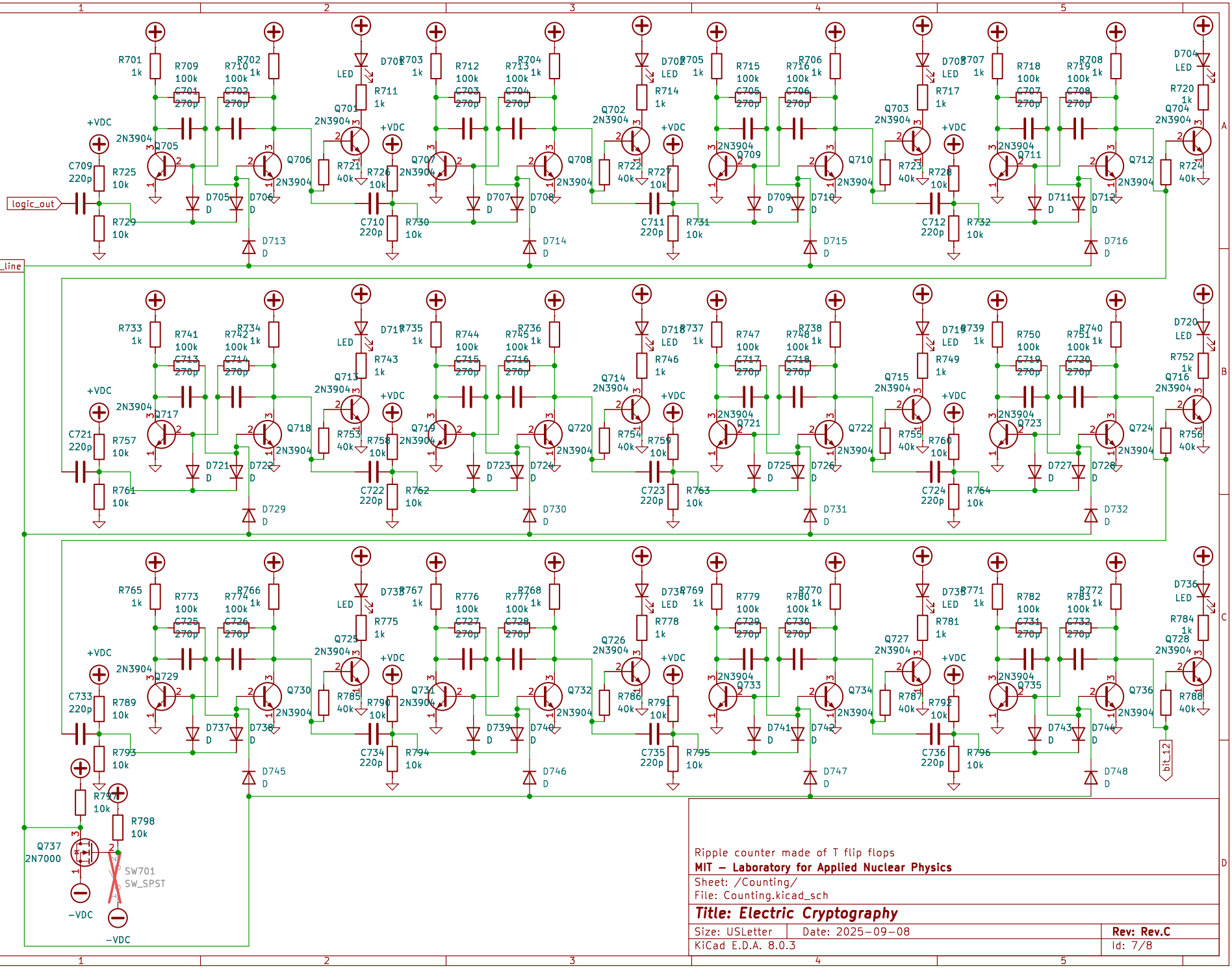
Sheet: /Shaping and Peak detection/

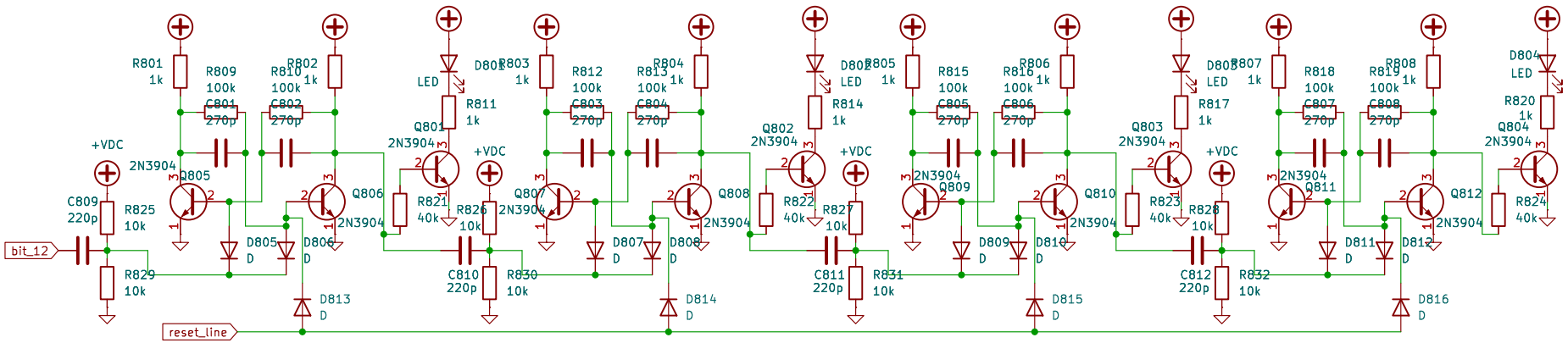
File: Shaping_Peakdet.kicad_sch

Title: Electric Cryptography

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peak_det_out
R604 1k
R605 1k
R603 1k
R606 15k
2N3906 Q603
Q604 2N3904
Q605 2N3904
R608 15k
R610 1k
R607 1k
TP601 TestPoint
RV601 R_Potentiometer_Trim
Q607Q608 2N3904
R616 10k
Q611 2N3904
R621 900
R622 1k
R620 1k
Q618 2N3904
C604 10n
R625 1k
R627 1k
R602 1k
Q602 2N7000
Q606 2N7000
R601 1k
Q601 2N7000
Holdoff
R609 1k
Q609 2N7000
peak_det_cap
Q612 2N7000
R623 20
R617 1k
R618 1k
C603 2.2n
R619 1k
Q616 2N3904
Q617 2N3904
R649 20k
Q630 2N3904
R624 1k
R611 1k
Q610 2N7000
logic_out
R612 1k
Q613 2N7000
R614 100k
C601 270p
R615 100k
C602 270p
R613 1k
Q614 2N3904
Q615 2N3904
D601 D
D602 D
R628 10k
R629 10k
SW_Push_DPDT SW601
R632 10k
-VDC
R626 5.6k
C605 10n
Q619 2N3904
R630 5.6k
R631 5.6k
R637 1k
C606 4.7n
R635 10k
R633 1k
Q621 2N3904
Q620 2N7000
R640 20
R636 1k
R638 1k
R634 1k
2N3906 Q622
R639 15k
Q625 2N3904
R643 15k
R645 1k
R642 1k
Q624 2N3904
Q626 2N3904
R644 22k
TP602 TestPoint
RV602 R_Potentiometer_Trim
Q627 2N3904
2N3904 Q628
R646 10k
Q629 2N3904
R647 900
R648 1k
ULD and LLD comparators and associated logic
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Sheet: /SCA and Logic/
File: SCA_Logic.kicad_sch
Title: Electric Cryptography
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logic_out
reset_line
bit_12
+VDC
-VDC
2N3904
2N7000
LED
SW701
SW_SPST
Ripple counter made of T flip flops
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Sheet: /Counting/
File: Counting.kicad_sch
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Ripple counter continued

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Sheet: /Counting_pt2/
File: counting_cont.kicad_sch

Title: Electric Cryptography

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KiCad E.D.A. 8.0.3		Id: 8/8